25ECPCC301IAE-I

Seat No. : FU125 7055

15 AUG 2026

Student Name : Harsh

St. John College of Engineering and Management, Palghar,
Maharashtra
Autonomous Institute

Internal Assessment Examinations, August 2026

B.Tech. Electronics & Computer Science - SEMESTER III

25ECPCC301 - Mathematics for ECS

Time : 1 Hour **Maximum Marks : 20**

St. John College of Engineering & Management ★ SJCEM PALGHAR

Instructions

- All Questions are compulsory and carry equal weightage of 5 marks
- Assume suitable data wherever required

Q1	Attempt any 5 out of 6 Questions	Max. marks(5)	Course Outcome	RBT Level
	a)What is truth table of NOT	1	CO1	L1
	b)Define injective function with an example.	1	CO2	L1
	c)Define Algebraic Multiplicity.	1	CO3	L1
	d)What is the Truth table for AND?	1	CO1	L1
	e)Define Symmetric Relation with suitable example.	1	CO2	L1
	f)When two vectors are Orthogonal.	1	CO3	L1
Q2	a)Prove by Mathematical Induction $P(n) = 1+2+3+\ldots+n = \frac{n(n+1)}{2}$	5	CO1	L3
	OR			
	b)Construct truth tables to check whether the statement $\neg(p \wedge q) \rightarrow (\neg p \vee \neg q)$ is tautology, contradictory or contingent.	5	CO1	L3
Q3	a)Let $A = \{1,2,3,4\}$. Let the relation R be defined on set A by " *if x is greater than y then xRy*". Find R, its matrix, its digraph and diagram.	5	CO2	L3
	OR			
	b)Let $A = \{1,2,3,4\}$ and $R = \{(1,1),(2,2),(3,3),(4,4),(2,1),(1,2),(2,3),(3,2),(3,1),(1,3)\}$. Prove that R is an equivalence relation. Also find its matrix and draw its digraph.	5	CO2	L3
Q4	a) Find all vectors in R^3 with the Euclidean norm that are orthogonal to $u_1 = (1,1,1)$, $u_2 = (1,1,0)$.	5	CO3	L3

	OR			
	b)Calculate the eigenvalues and eigenvectors of $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.	5	CO3	L3

